

Solution Architecture

Date 30 September 2026

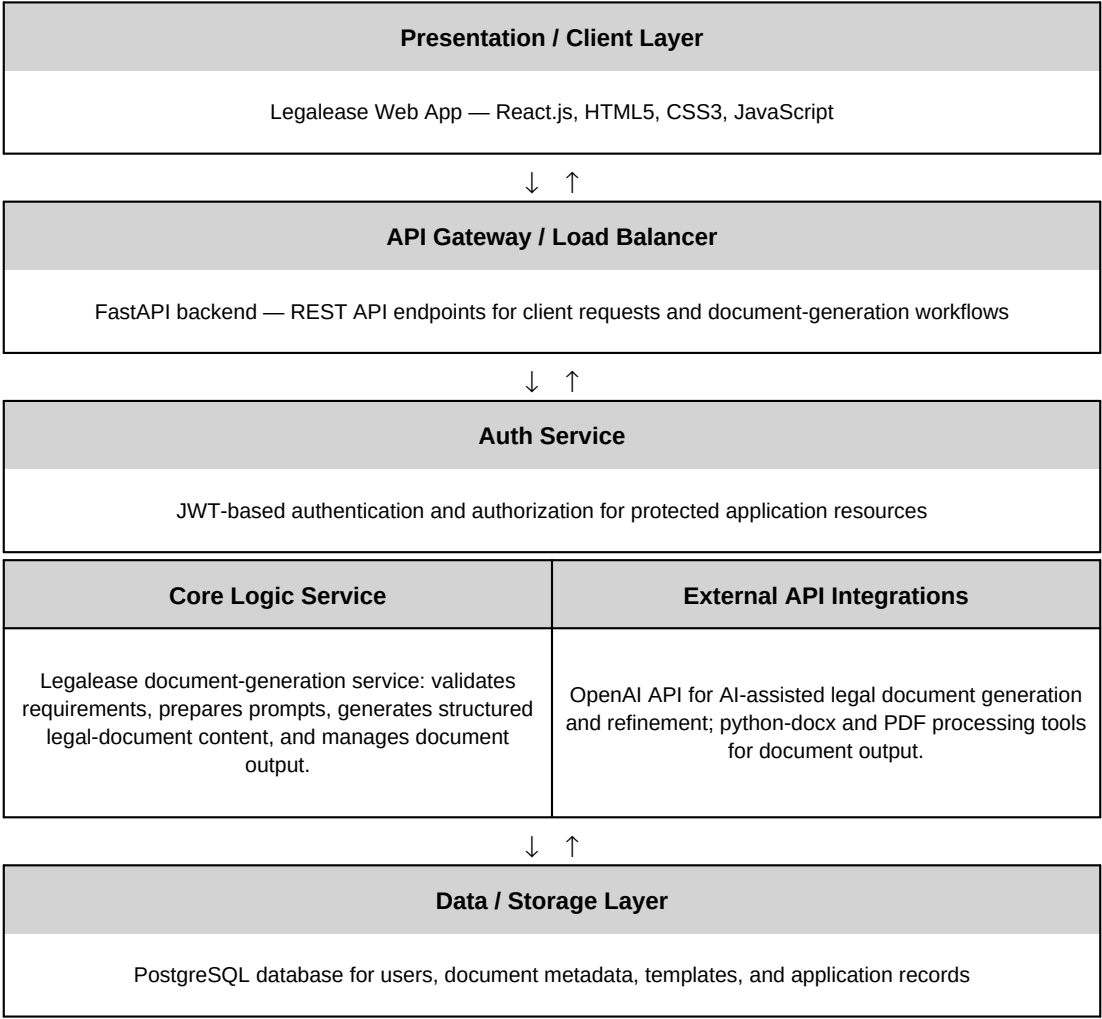
Team ID SWTID-2026-5550

Project Name Legalease: AI-powered legal document generator

Maximum Marks 5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below to map out your Presentation Layer, Application Layer, and Data Layer. The technologies and components below are selected for the Legalease solution.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering legal-document requirements, reviewing generated content, and interacting with the Legalease application.	React.js, HTML5, CSS3, JavaScript
API Gateway	Receives client requests, routes them to backend services, and exposes APIs for authentication, document generation, and document management.	Python, FastAPI, REST API
Microservices	Handles authentication, requirement processing, AI-assisted document generation, document formatting, validation, and generation of editable/output documents.	FastAPI, Python, OpenAI API, python-docx, PDF processing libraries
Database	Stores application data such as users, document metadata, templates, and generated-document records.	PostgreSQL

Architecture Flow: User → React.js Presentation Layer → FastAPI API Layer → Authentication / Core Logic → OpenAI API and document-processing tools → PostgreSQL Data Layer.

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